MIT-BIH Arrhythmia Database

Analyzing Abnormal Heartbeats

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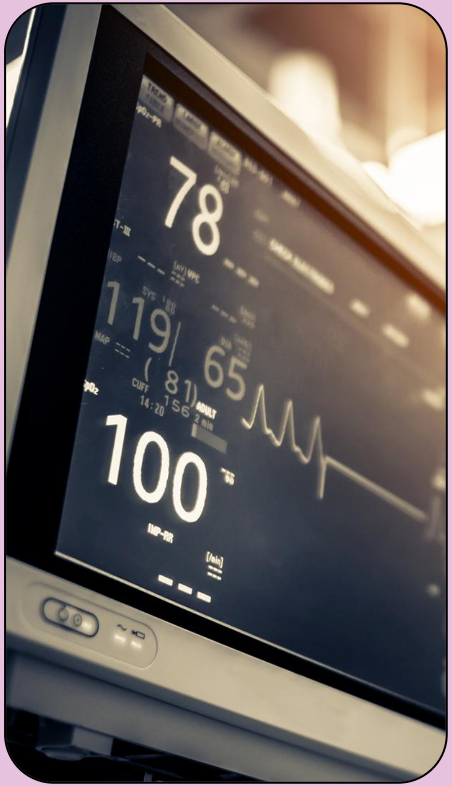
INTRODUCTION

- MIT-BIH Arrhythmia-Database through PhysioNet (1999)
- Boston's Beth Israel Hospital in collaboration with MIT
- Data includes 48 half-hour two-channel ECG recordings
    - 47 individuals (1975-1979)
    - Sampled at 360 Hz
    - 2 cardiologists
    - 110,000 heartbeats total
    - No missing data!
        - Research-grade
- Looking at ventricular beats:
    - **Transitions** between **normal** and **ventricular** (abnormal) beats
    - **Variability** in **timing** of the heartbeats

The Data

- We chose record 100 and 208
  - Record 100 = 2239 normal + 1 V beats + 33 A beats
  - Record 208 = 1586 normal + 992 V beats + 0 A beats
- Data includes 4 file types
  - .dat: Binary-encoded ECG waveforms
    - Record 100 = MLII, V5
    - Record 208 = MLII, V1
  - .atr: timing and type of heartbeat with symbolic codes
  - .hea: text file to explain heartbeat with channeling information
  - .xws: auxiliary file that contains waveform display settings



Data Extraction

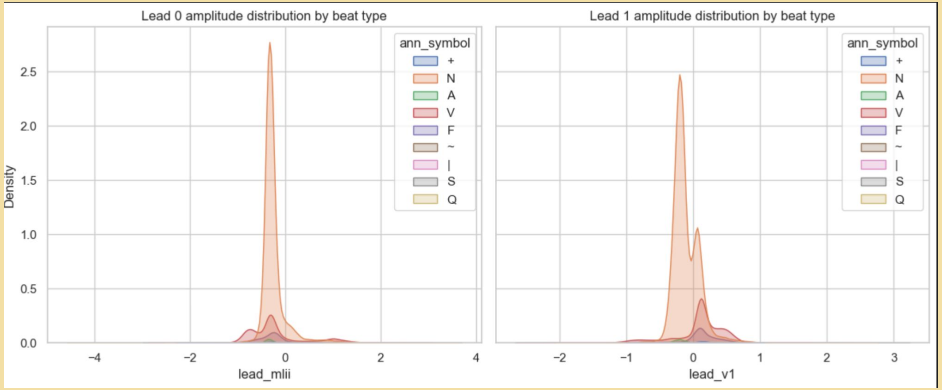
Exploratory Data Analysis

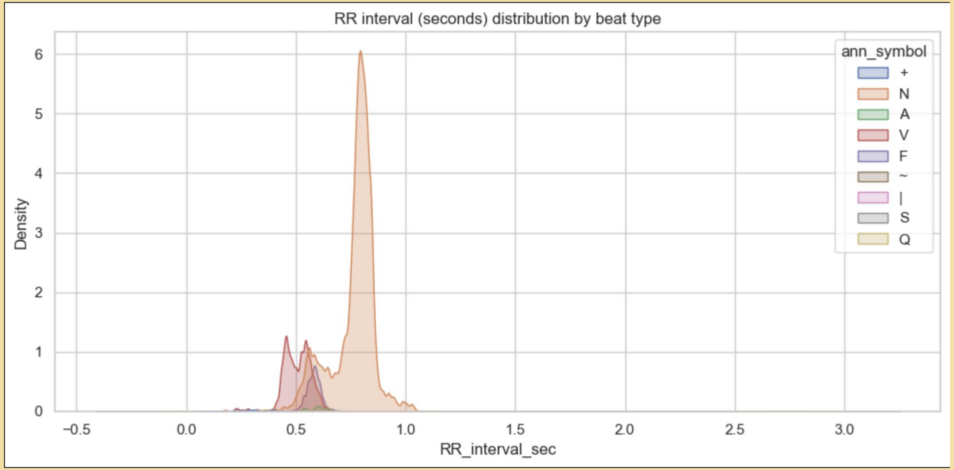
1. Class Imbalance

- Our dataset contains far more normal heartbeats (N) than ventricular beats (V)
- This reflects reality: most heartbeats are normal, even in patients with arrhythmias
- Challenge: We'll need to account for this imbalance when training our model to avoid it simply predicting "normal" for everything

2. Timing Differences

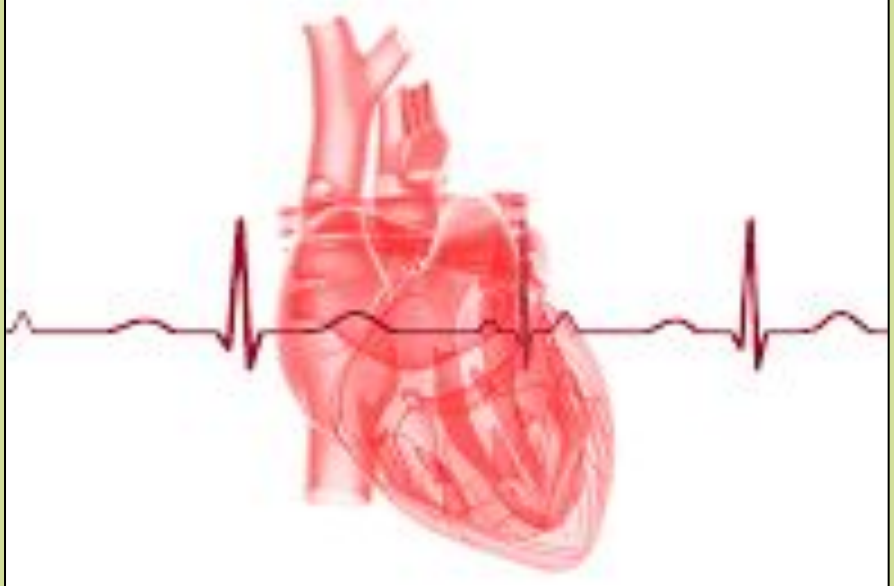
- Normal beats follow a regular rhythm with RR intervals around 0.8-1.0 seconds (equivalent to 60-75 beats per minute)
- PVCs arrive early: their RR intervals cluster around 0.4-0.6 seconds
    a. PVCs are "premature" contractions that disrupt the normal rhythm
- The RR interval = a powerful predictor
    a. if a beat arrives much earlier than expected, it's likely a PVC





Non-Parametric Model

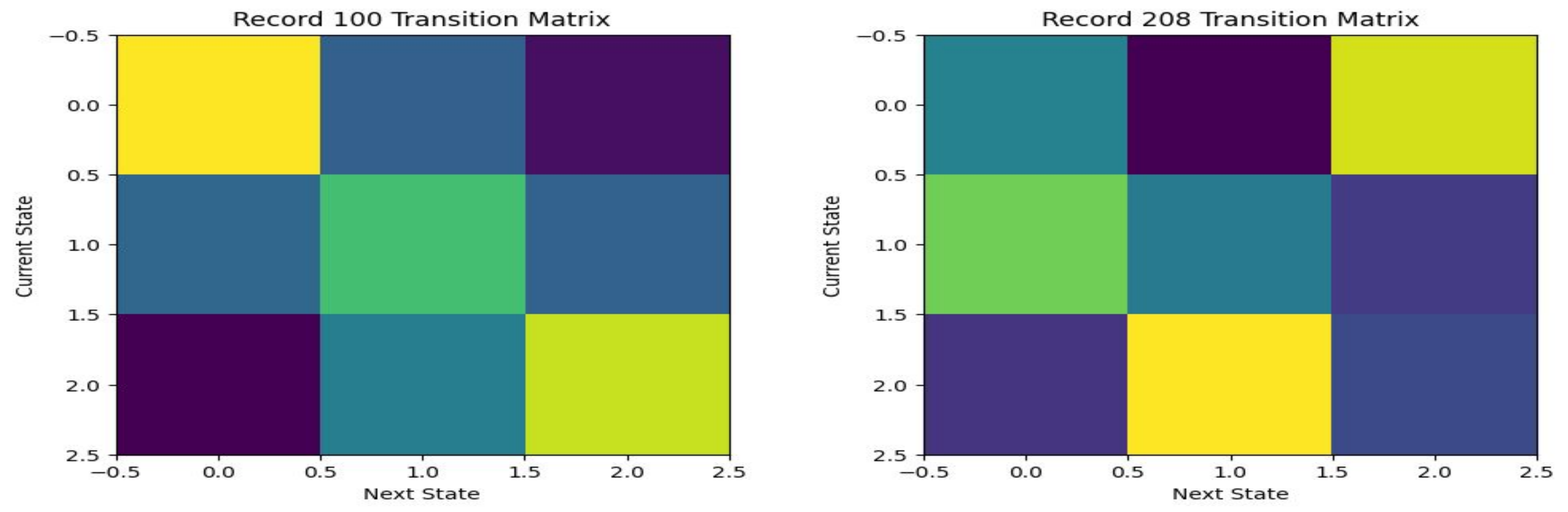
- **Our Goal:** Analyze and predict a PVC from a regular heartbeat.
- **Our Model** –> Random Forest Classifier:
  - **Why**:
    - Captures Nonlinear patterns without assumptions
    - Handles noise and class imbalance
    - High Accuracy
  - **Alternatives**: Logistic Regression, KNN, CNN
  - We trained a Random Forest model using both timing (RR intervals) and shape (ECG amplitude) features to distinguish normal beats from PVCs.



Method 1: Markov Chain

Transition Matrices

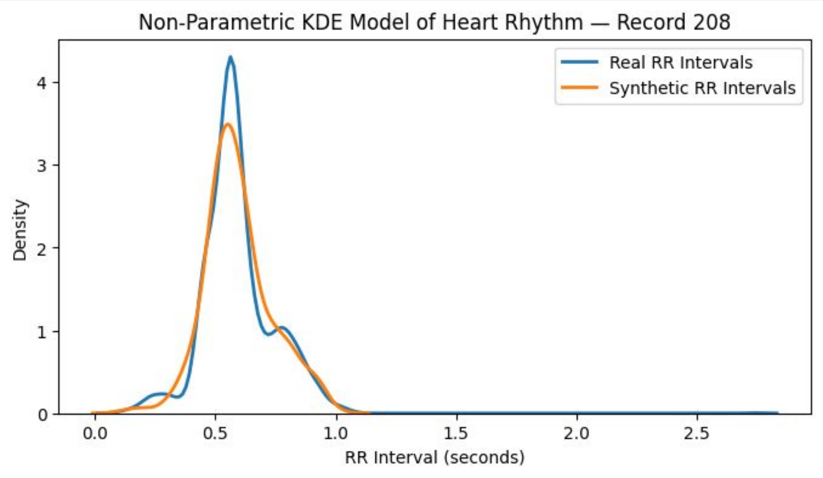
Markov transition models: Capturing how the heart moves between different states (normal → PVC → back to normal)

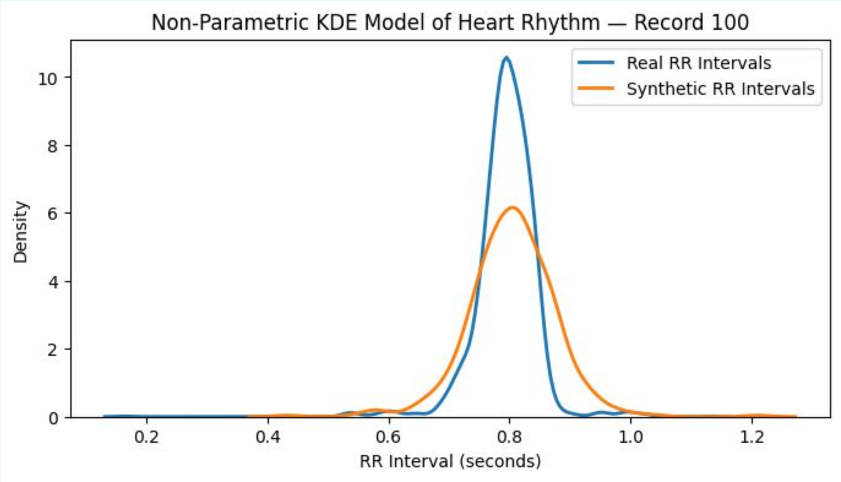


Method 2: KDE

KDE Plot

KDE: plots the PDF and show where RR interval values are concentrated, for 500 synthetic samples.
Results: Synthetic and Actual RR Intervals follow same shape with strong peaks at 0.5 - 0.6 (100 -120 BPM) and 0.8 (75 BPM) -> steady and natural heartbeat

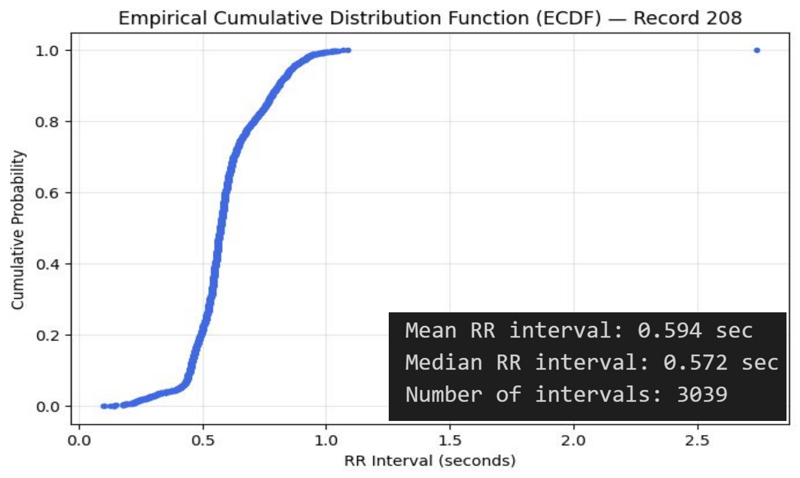


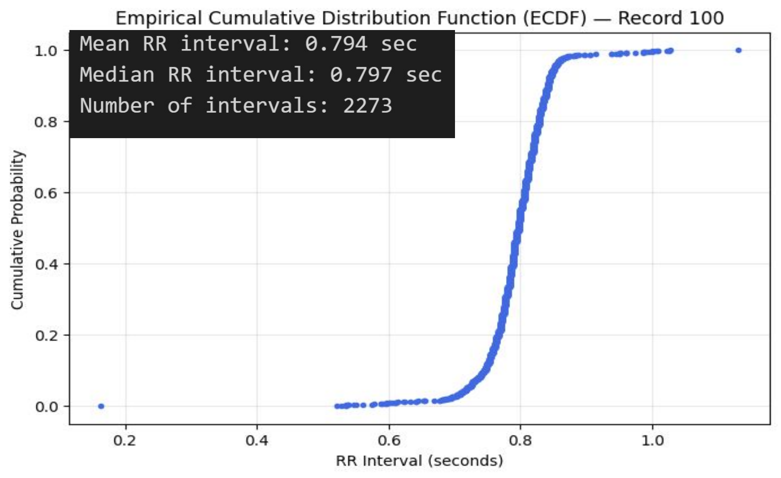


Method 3: ECDF

ECDF Plot

ECDF: tells us the proportion of observations less than or equal to a given data value.
Results: Both ECDF curves are steep, indicating tight range -> heart beats are consistent and steady





Critical Evaluation

Results

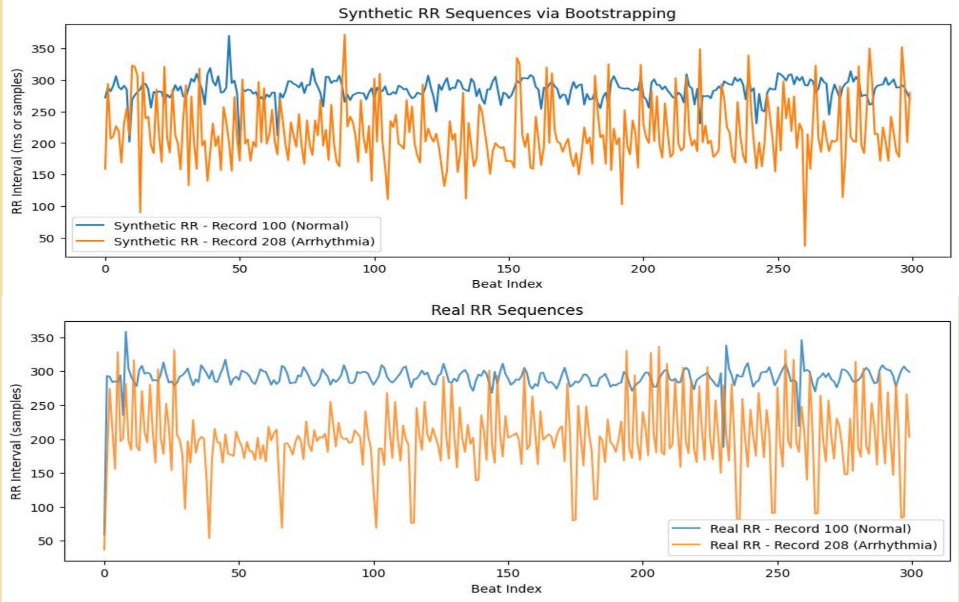
- Synthetic RR sequences vs Real RR sequences
- Bootstrapping did NOT capture underlying issues
    - Temporal dependencies
    - Rhythm patterns

Differences:

- Overlap
- Irregularity
- Less separation

Uncertainty:

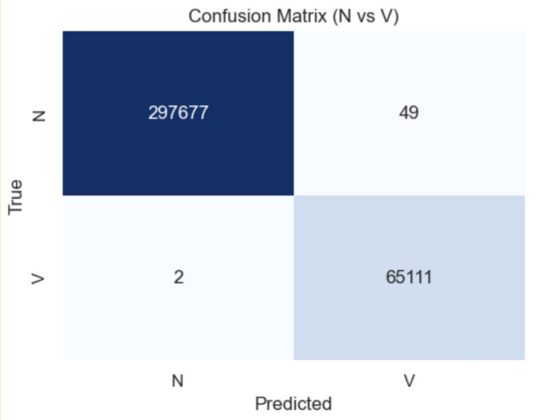
- Uncertainty is present
- Less representative and robust

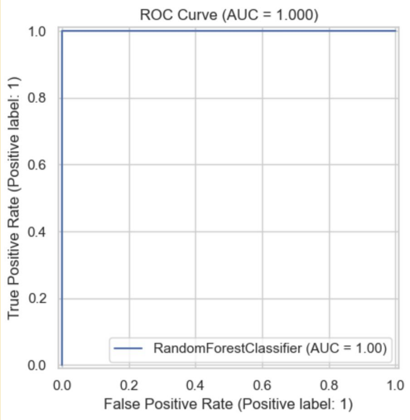


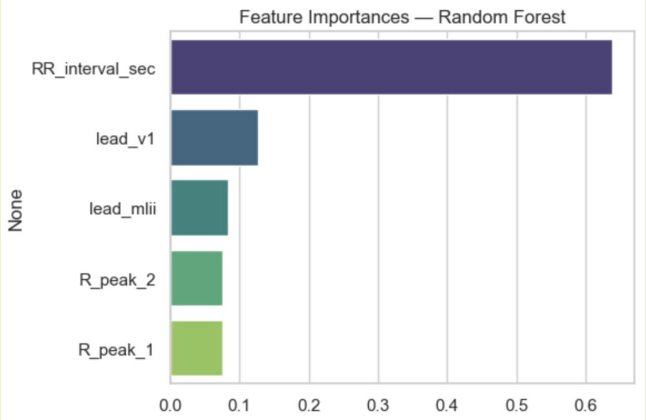
Results

Modeling Process

- **Feature Importance:**
  - **RR_interval_sec** (Timing): ***By far the most important feature*** (~60% importance)
    - Confirms our EDA finding: PVCs arrive early, and this timing signature is the strongest predictor
  - **lead_v1** (Shape from ECG Lead 1): Second most important (~12% importance)
    - The electrical signal shape also helps distinguish PVCs
  - **lead_mlii** (Shape from Modified Lead II): Third (~10% importance)
  - **R_peak amplitudes**: Contribute modestly to prediction
- **Conclusion:** Our model can reliably detect PVCs in ECG data with near-perfect accuracy.







Results, Limitations, Future Work

Conclusion

Results

- Normal rhythm (Record 100): Synthetic closely matched natural.
- Arrhythmic rhythm (Record 208): Synthetic didn’t match natural.

Limitations

- The first-order Markov model only considers one previous beat, missing longer-term patterns.
- Biggest challenge for KDE and ECDF is choosing the sample size - if too small not representative of true shape and if too large distribution is too smooth

Future Work

- Use a higher-order Markov model so predictions depend on multiple prior beats.
- Adding hidden states to represent physiological conditions
- KDE and ECDF match underlying distribution, meaning there is no need to adjust